

Git & GitHub

- **Create a GitHub account**

Step 1: Open GitHub

Go to:

GitHub

Step 2: Click Sign up

Click the Sign up button in the top-right corner.

Step 3: Enter Your Email

Enter your email address and click Continue.

Step 4: Create a Password

Create a strong password and click Continue.

Step 5: Choose a Username

Ex khadeer329

Step 6: Complete the Verification

Step 7: Verify Your Email

Step 10: Create Your First Repository

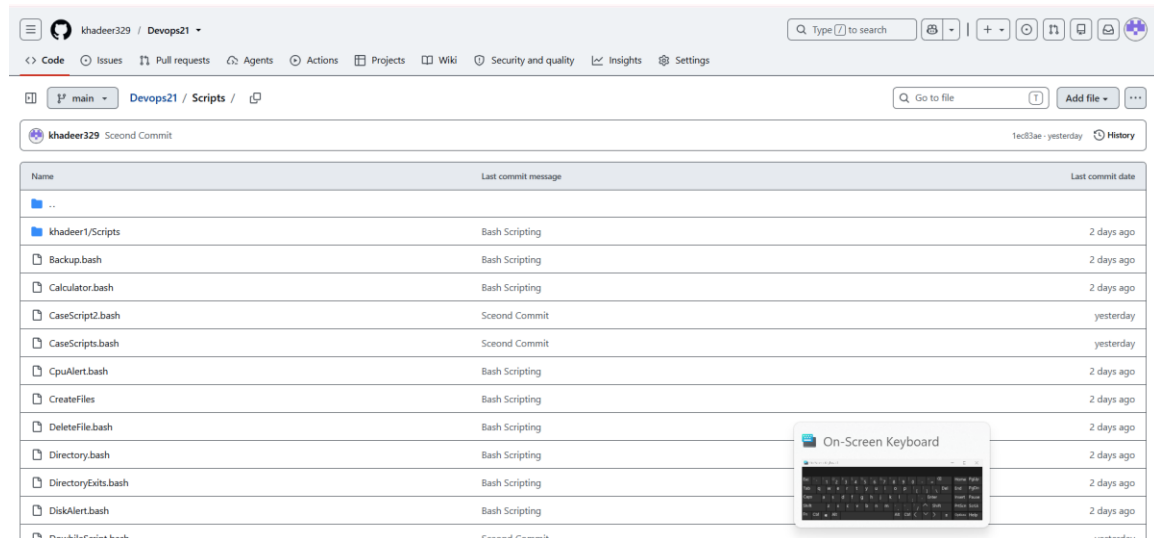
After logging in:

Click the + icon (top-right).

Select New repository.

Repository name:

Devops21



- **Required Steps: Set Up MFA (Multi-Factor Authentication) for GitHub Using a Mobile App**

Step 1: Sign in to GitHub

Go to:

GitHub

Log in to your account.

Step 2: Open Settings

Click your profile picture (top-right).

Click Settings.

Step 3: Open Password and Authentication

In the left menu, click:

Password and authentication

Step 4: Start MFA Setup

Scroll to Two-factor authentication.

Click: Enable two-factor authentication

Step 5: Choose Authentication App

Select:

Authentication App

Step 6: Install an Authenticator App on Your Phone

Install any one of these:

Google Authenticator

Step 7: Scan the QR Code

GitHub will display a QR code.

On your phone:

Tap + (Add Account).

Choose Scan QR Code.

Scan the QR code shown on GitHub.

Step 8: Enter the Verification Code

The app will generate a 6-digit code.

Enter that code into GitHub and click:

Continue or Enable

Step 9: Verify MFA

- Sign out of GitHub.
- Sign in again.
- Enter:
- Your password.
- The 6-digit code from your authenticator app

3. Git Basic Commands

1. Initialize Local Git Repository

Commands: git init, ll, ls -a.

Initializes an empty repository and verifies the hidden .git folder.

```
DELL@KHADEER MINGW64 ~/Desktop/GitPractices
$ git init
Initialized empty Git repository in C:/Users/DELL/Desktop/GitPractices/.git/

DELL@KHADEER MINGW64 ~/Desktop/GitPractices (master)
$ ll
total 0

DELL@KHADEER MINGW64 ~/Desktop/GitPractices (master)
$ ls -a
./ ../ .git/
```

2. Stage Dockerfile and Verify Status

Commands: git add dockerfile, git status.

```
DELL@KHADEER MINGW64 ~/Desktop/GitPractices (master)
$ git add dockerfile

DELL@KHADEER MINGW64 ~/Desktop/GitPractices (master)
$ git stauts
git: 'stauts' is not a git command. See 'git --help'.

The most similar command is
    status

DELL@KHADEER MINGW64 ~/Desktop/GitPractices (master)
$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   dockerfile
```

3. Create First Commit

Commands: git commit -m "my first commit", git status, touch a b c d e f g h j.

```
DELL@KHADEER MINGW64 ~/Desktop/GitPractices (master)
$ git commit -m "my first commit"
[master (root-commit) 96287d8] my first commit
1 file changed, 30 insertions(+)
create mode 100644 dockerfile

DELL@KHADEER MINGW64 ~/Desktop/GitPractices (master)
$ git status
On branch master
nothing to commit, working tree clean

DELL@KHADEER MINGW64 ~/Desktop/GitPractices (master)
$ touch a b c d e f g h j
```

```

DELL@KHADEER MINGW64 ~/Desktop/GitPractices (master)
$ touch a b c d e f g h j

DELL@KHADEER MINGW64 ~/Desktop/GitPractices (master)
$ ll
total 1
-rw-r--r-- 1 DELL 197121 0 Jul 7 15:07 a
-rw-r--r-- 1 DELL 197121 0 Jul 7 15:07 b
-rw-r--r-- 1 DELL 197121 0 Jul 7 15:07 c
-rw-r--r-- 1 DELL 197121 0 Jul 7 15:07 d
-rw-r--r-- 1 DELL 197121 581 Jul 7 15:04 dockerfile
-rw-r--r-- 1 DELL 197121 0 Jul 7 15:07 e
-rw-r--r-- 1 DELL 197121 0 Jul 7 15:07 f
-rw-r--r-- 1 DELL 197121 0 Jul 7 15:07 g
-rw-r--r-- 1 DELL 197121 0 Jul 7 15:07 h
-rw-r--r-- 1 DELL 197121 0 Jul 7 15:07 j

DELL@KHADEER MINGW64 ~/Desktop/GitPractices (master)
$ git status
On branch master
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        a
        b
        c
        d
        e
        f
        g
        h
        j

nothing added to commit but untracked files present (use "git add" to track)

```

4. Stage All Files and Create Second Commit

Commands: git add ., git status, git commit -m "my Second commit".

```
DELL@KHADEER MINGW64 ~/Desktop/GitPractices (master)
$ git add .

DELL@KHADEER MINGW64 ~/Desktop/GitPractices (master)
$ git status
On branch master
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   a
        new file:   b
        new file:   c
        new file:   d
        new file:   e
        new file:   f
        new file:   g
        new file:   h
        new file:   j

DELL@KHADEER MINGW64 ~/Desktop/GitPractices (master)
$ git commit -m " my Second commit"
[master e9592fa] my Second commit
9 files changed, 0 insertions(+), 0 deletions(-)
create mode 100644 a
create mode 100644 b
create mode 100644 c
create mode 100644 d
create mode 100644 e
create mode 100644 f
create mode 100644 g
create mode 100644 h
create mode 100644 j
```

5. View Commit History and Git Configuration

Commands: git log, git config --list, git config user.name, git config user.email.

```
DELL@KHADEER MINGW64 ~/Desktop/GitPractices (master)
$ git log
commit e9592fadd903ebffca833aafc784afb35fcac160 (HEAD -> master)
Author: khadeer9 <khadeershaik0329@gmail.com>
Date: Tue Jul 7 15:08:59 2026 +0530

    my Second commit

commit 96287d82d3a1eef459b3fd12a1b6c770fb8427de
Author: khadeer9 <khadeershaik0329@gmail.com>
Date: Tue Jul 7 15:06:33 2026 +0530

    my first commit
```

```
DELL@KHADEER MINGW64 ~/Desktop/GitPractices (master)
$ git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=openssl
http.sslcainfo=C:/Program Files/Git/mingw64/etc/ssl/certs/ca-bundle.crt
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.rebase=false
credential.helper=manager
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=master
user.name=khadeer9
user.email=khadeershaik0329@gmail.com
core.repositoryformatversion=0
core.filemode=false
core.bare=false
core.logallrefupdates=true
core.symlinks=false
core.ignorecase=true
```

6. Create GitHub Repository

Create a new repository on GitHub and note the SSH repository URL.

Quick setup — if you've done this kind of thing before

 Set up in Desktop or ☐ HTTPS ☐ SSH 

Get started by [creating a new file](#) or [uploading an existing file](#). We recommend every repository include a [README](#), [LICENSE](#), and [.gitignore](#).

...or create a new repository on the command line

```
echo "# GitDemo" >> README.md
git init
git add README.md
git commit -m "first commit"
git branch -M main
git remote add origin git@github.com:khadeer329/GitDemo.git
git push -u origin main
```



7. Configure Remote Repository

Commands: git push (fails because no remote), git remote add origin..., git remote -v.

```
DELL@KHADEER MINGW64 ~/Desktop/GitPractices (master)
$ git config user.name
khadeer329

DELL@KHADEER MINGW64 ~/Desktop/GitPractices (master)
$ git config user.email
khadeershaik0329@gmail.com
```

8. Generate SSH Key

Commands: ssh-keygen and cat ~/.ssh/id_ed25519.pub.

```

DELL@KHADEER MINGW64 ~/Desktop/GitPractices (master)
$ ssh-keygen
Generating public/private ed25519 key pair.
Enter file in which to save the key (/c/Users/DELL/.ssh/id_ed25519):
Enter passphrase for "/c/Users/DELL/.ssh/id_ed25519" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /c/Users/DELL/.ssh/id_ed25519
Your public key has been saved in /c/Users/DELL/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:tFdVF/DiFG7kiaii/ncsLRY7K7PLqZ1HnmdesdsUB6g DELL@KHADEER
The key's randomart image is:
+--[ED25519 256]--+
|      .+oo+      |
|    . *.+ .    |
|  .. o.O .    |
| . . . . + o   |
|  . .SE.. o .  |
| . .O . .O O   |
| . .O * .O .   |
| . ooo@ * . +   |
| ooOO.X. . .   |
+-----[SHA256]-----+

```

```

DELL@KHADEER MINGW64 ~/Desktop/GitPractices (master)
$ cat /c/Users/DELL/.ssh/id_ed25519.pub

```

9. Add SSH Key to GitHub

Paste the copied public key into GitHub > Settings > SSH and GPG Keys.

Title

gitdemo

Key type

Authentication Key ↕

Key

ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIGY8af0Q77CUdTyhVjE0RYa2jfcSEZZRTRfjZSVxj+j DELL@KHADEER|

Add SSH key

SSH keys

[New SSH key](#)

This is a list of SSH keys associated with your account. Remove any keys that you do not recognize.

Authentication keys



gitdemo
SHA256:tFdVF/diF67kiaii/ncsLRY7K7PLqZ1HnmdeSDsUB6g
Added on Jul 7, 2026
Never used — Read/write

Delete

10. Push Repository to GitHub

Command: `git push -u origin master`.

```
DELL@KHADEER MINGW64 ~/Desktop/GitPractices (master)
$ git push -u origin master
The authenticity of host 'github.com (20.207.73.82)' can't be established.
ED25519 key fingerprint is: SHA256:+DiY3wvV6TuJJhbpZisF/zLDA0zPMSvHdkr4UvCOqU
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added 'github.com' (ED25519) to the list of known hosts.
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 8 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (6/6), 862 bytes | 215.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:khadeer329/GitDemo.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.
```

11. Verify Repository on GitHub

Repository now contains dockerfile and files a–j.

master 1 Branch 0 Tags

[Add file](#)

[Code](#)

khadeer329 my Second commit	e9592fa · 43 minutes ago	2 Commits
a	my Second commit	43 minutes ago
b	my Second commit	43 minutes ago
c	my Second commit	43 minutes ago
d	my Second commit	43 minutes ago
dockerfile	my first commit	45 minutes ago
e	my Second commit	43 minutes ago
f	my Second commit	43 minutes ago
g	my Second commit	43 minutes ago
h	my Second commit	43 minutes ago
j	my Second commit	43 minutes ago

12. Create Additional Commit and Verify

After creating an additional file and pushing, verify the latest commit on GitHub.

master 1 Branch 0 Tags

Go to file

Add file

Code

khadeer329	Create khadeer	829e7b5 · now	3 Commits
a	my Second commit	46 minutes ago	
b	my Second commit	46 minutes ago	
c	my Second commit	46 minutes ago	
d	my Second commit	46 minutes ago	
dockerfile	my first commit	48 minutes ago	
e	my Second commit	46 minutes ago	
f	my Second commit	46 minutes ago	
g	my Second commit	46 minutes ago	
h	my Second commit	46 minutes ago	
j	my Second commit	46 minutes ago	
khadeer	Create khadeer	now	

13. Pull, Push and Final Log

Commands: git pull, git push, git log.

```
DELL@KHADEER MINGW64 ~/Desktop/GitPractices (master)
$ git pull
Already up to date.

DELL@KHADEER MINGW64 ~/Desktop/GitPractices (master)
$ git push
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 8 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 460 bytes | 230.00 KiB/s, done.
Total 4 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 1 local object.
To github.com:khadeer329/GitDemo.git
829e7b5..a709234 master -> master
```

14. Revert

Commands: git revert, git log.

```
DELL@KHADEER MINGW64 ~/Desktop/GitPractices (master)
$ git log
commit 8400c48ffd968bcfc7ee50e96680ff8a400491db (HEAD -> master)
Author: khadeer329 <khadeershaik0329@gmail.com>
Date: Tue Jul 7 16:08:10 2026 +0530

    Revert "add abc file"

    This reverts commit 60a1c8c4c9816323abf67b1f48be5895aef5a9a5.

commit a709234ff7246e6c0b77b610c6eb84eae5d9c705 (origin/master, origin/HEAD)
Merge: 60a1c8c 829e7b5
Author: khadeer329 <khadeershaik0329@gmail.com>
Date: Tue Jul 7 15:56:57 2026 +0530

    Merge branch 'master' of github.com:khadeer329/GitDemo

commit 60a1c8c4c9816323abf67b1f48be5895aef5a9a5
Author: khadeer329 <khadeershaik0329@gmail.com>
Date: Tue Jul 7 15:56:25 2026 +0530

    add abc file

commit 829e7b562f63ff76728e0d4ab93bbf89fe176a56
Author: khadeer329 <155939706+khadeer329@users.noreply.github.com>
Date: Tue Jul 7 15:54:52 2026 +0530

    Create khadeer

commit e9592fadd903ebffca833aafc784afb35fcac160
Author: khadeer9 <khadeershaik0329@gmail.com>
Date: Tue Jul 7 15:08:59 2026 +0530

    my Second commit

commit 96287d82d3a1eef459b3fd12a1b6c770fb8427de
Author: khadeer9 <khadeershaik0329@gmail.com>
Date: Tue Jul 7 15:06:33 2026 +0530

    my first commit
```